

MENTORA ACADEMY

EXCELLENCE IN EDUCATION

Edexcel International A Level Chemistry (WCH14)

UNIT 4 — TOPICAL WORKSHEET PACK: 14A.2

HYDROGEN ION CONCENTRATION AND THE PH SCALE

CANDIDATE NAME:	Usman
SUBJECT & CODE:	Chemistry Unit 4 — Rates, Equilibria & Organic (WCH14)
TOPIC AREA:	14: Acid-Base Equilibria
SUB-TOPIC:	14A.2: Hydrogen Ion Concentration and the pH Scale
TOTAL QUESTIONS:	1 Questions (Structured + MCQs + Calculations)
TOTAL MARKS:	1 Marks
EXAMINER FAQs:	1 Empirical Examiner FAQs & Traps Included

HOW TO USE THIS TOPICAL PACK:

1. Complete all structured questions under timed exam conditions.
2. Pay close attention to command words (Explain, Deduce, Calculate, State).
3. Review the complete worked mark scheme in Section C.
4. Study the Empirical Examiner FAQs in Section D to avoid common candidate traps.

SECTION A: TOPICAL PRACTICE QUESTIONS — 14A.2

Q1. Calculate the pH of 0.150 mol dm⁻³ HCl.

[1 Marks]

WCH14/01/Jun21/Q4(a)

SECTION B: WORKED MARK SCHEME — 14A.2

Q	MARKING SCHEME & ACCEPTABLE ANSWERS	MARKS
Q1	Q1: Calculate the pH of 0.150 mol dm ⁻³ HCl. pH = $-\log(0.150) = 0.82$	[1]

SECTION C: EMPIRICAL EXAMINER FAQS & COMMON TRAPS — 14A.2

FAQ #1: Do I need to worry about dibasic acids like H₂SO₄?**Category: Examiner Trap & Technique****COMMON EXAMINER TRAP / MISTAKE:**

Do I need to worry about dibasic acids like H₂SO₄?

MODEL ANSWER / EXAMINER EXPECTATION:

Yes, for a strong dibasic acid, [H⁺] is approximately twice the concentration of the acid. Remember to multiply by 2 before taking the -log.